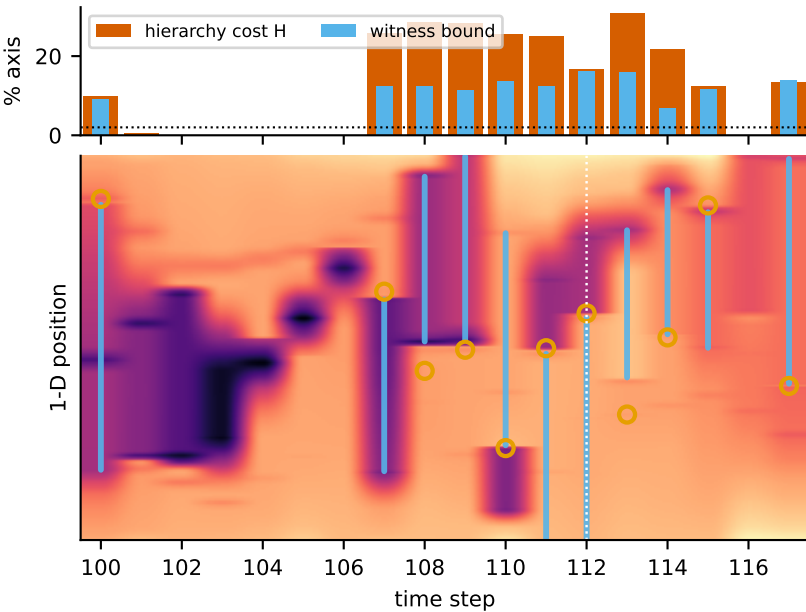
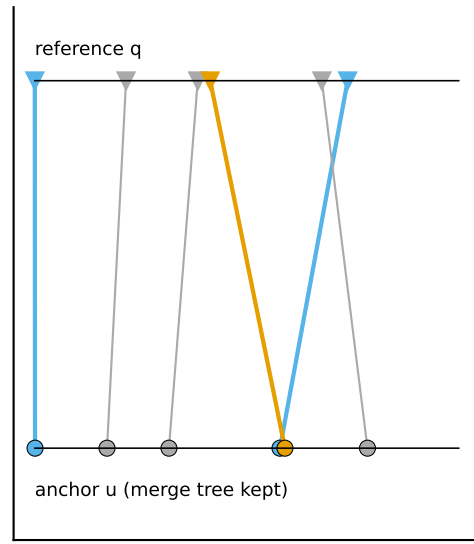


(a) merge tree kept: certificate and witness triples



(b) step 112: witness bound 16%,  $\tau^* = 17\%$   
blue: one subtree; orange: lies between them in q



(c) relaxed ( $\kappa = 20\%$ ), optimal filling;  $\blacktriangledown$  relaxed steps

